Thesis Presentation:

Forecasting with Linear Augmented Information Combination

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EBS Honours/Masters Presentation

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Outline

Part 1. *Research Question and Method*

1.1 Forecast reconciliation

1.2 Multivariate regression framework with flexible predictor set

Forecasting with **L**inear **A**ugmented **I**nformation **C**ombination

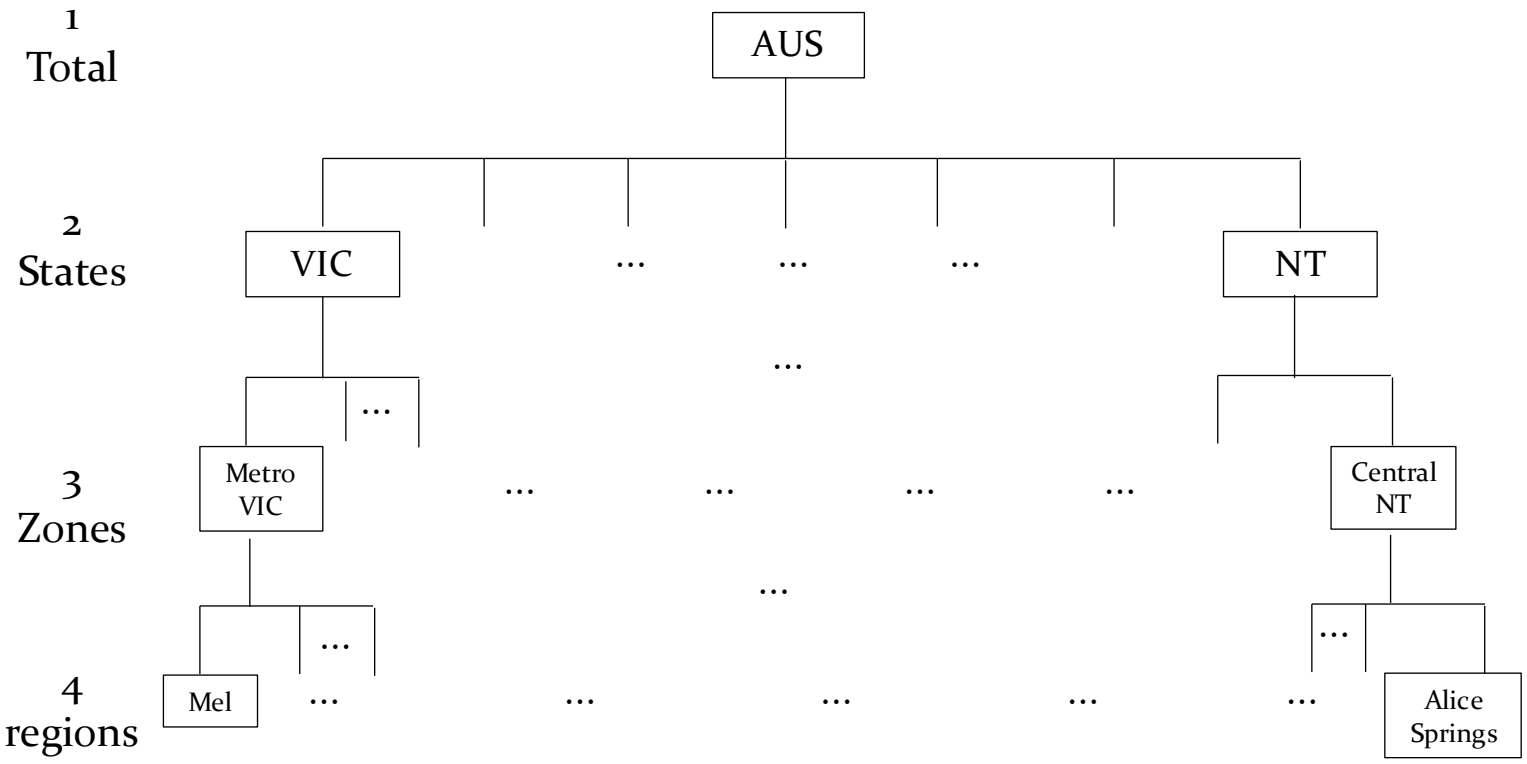
1.3 Linear components augmented (FLAP idea) in IComb predictor set

Part 2. *Empirical Investigation*

1.1 What is forecast reconciliation problem?

Levels

Domestic tourism hierarchy *(n=112 series)*



Hierarchical structure of data

$$\mathbf{y}_t = \mathbf{S}\mathbf{b}_t$$

Aggregation Constraints

$$\begin{bmatrix}\mathbf{a}_t\\ \mathbf{b}_t\end{bmatrix} = \begin{bmatrix}\mathbf{A}\\ \mathbf{I}_{n_b}\end{bmatrix}\mathbf{b}_t$$

$$\mathbf{C}\,\mathbf{y}_t = [\mathbf{I}_{n_a} \;\; -\mathbf{A}]\,\mathbf{y}_t = \mathbf{0}$$

Forecasting problem:

Base forecasts (h-step ahead) using

$$\mathfrak{T}_T = (\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T)$$

are *incoherent*

$$\hat{\mathbf{y}}_{T+h|t} \neq \mathbf{S}\,\hat{\mathbf{b}}_{T+h|t}$$

Base forecasts $\hat{\mathbf{y}}_{T+h|t}$ ⟶ Reconciliation ⟶ Coherent $\tilde{\mathbf{y}}_{T+h|t}$

1.2 How has the literature approached reconciliation?

Unbiasedness of base forecasts assumption:

$$\mathrm{E}\left[\hat{\mathbf{y}}_{\mathrm{T}+h|\mathrm{T}}|\mathfrak{T}_{\mathrm{T}}\right] = \mathrm{E}\left[\mathbf{y}_{\mathrm{T}+\mathrm{h}}|\mathfrak{T}_{\mathrm{T}}\right] \quad \Longrightarrow \quad \hat{\mathbf{y}}_{\mathrm{T}+h|\mathrm{T}} = \mathrm{E}\left[\mathbf{y}_{\mathrm{T}+\mathrm{h}}|\mathfrak{T}_{\mathrm{T}}\right] + \boldsymbol{\epsilon}, \qquad \mathrm{E}\left[\boldsymbol{\epsilon}|\mathfrak{T}_{\mathrm{T}}\right] = 0$$

Strong requirement (Nguyen et al. 2025)

(assumes base forecasts embody ***all*** useful information in the history of the hierarchy)

Minimum Trace (MinT) (Wickramasuriya et al., 2019)

- Unbiasedness assumption + Optimisation
- Find optimal $\mathbf{G}_h$ for:

$$\tilde{\mathbf{y}}_{\mathrm{T}+h|\mathrm{T}} = \mathbf{S}\,\tilde{\mathbf{b}}_{\mathrm{T}+h|\mathrm{T}} = \mathbf{S}\,\mathbf{G}_h\,\hat{\mathbf{y}}_{\mathrm{T}+h|\mathrm{T}}$$

- Solution is: $\mathbf{G}_{h,\mathrm{MinT}} = \left(\mathbf{S}'\mathbf{W}_h^{-1}\mathbf{S}\right)^{-1}\mathbf{S}'\mathbf{W}_h^{-1}$

where: $\mathbf{W}_h = \mathrm{Var}(\hat{\mathbf{e}}_{\mathrm{t}+h|\mathrm{t}}), \quad \hat{\mathbf{e}}_{\mathrm{t}+h|\mathrm{t}} = \mathbf{y}_{\mathrm{t}+\mathrm{h}} - \hat{\mathbf{y}}_{t+h|t}$

Information Combination (IComb) (Nguyen et al. 2025)

- Multivariate regression: $\mathbf{Y} = \mathbf{X}\,\mathbf{B} + \mathbf{E}$

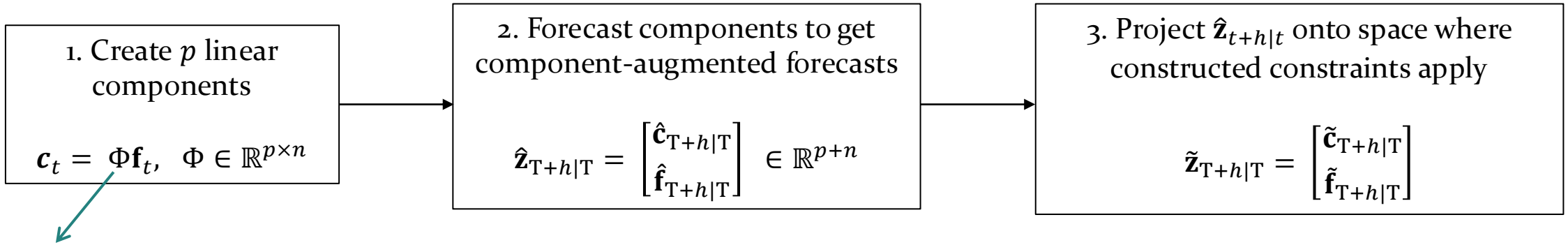
where: $\mathbf{Y}$: coherent realised hierarchy

$\mathbf{X}$: *any* predictor set, hierarchy in-sample forecasts

- $\tilde{\mathbf{y}}_{\mathrm{T}+h|\mathrm{T}} = \hat{\mathbf{B}}'\,\hat{\mathbf{y}}_{\mathrm{T}+h|\mathrm{T}}$ coherent as: $\mathbf{C}\,\hat{\mathbf{B}}' = \mathbf{C}\,\mathbf{Y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{0}$
- Regularisation:
  - IComb-S: Ridge (shrinkage) ↔ dense models
  - IComb-L: Multivariate lasso ↔ sparse models

1.3 Gap: how to expand the predictor set beyond forecasts of $\mathbf{y}_t$?

- Forecast Linear Augmented Projection (FLAP) (Yang et al., 2024) says you can improve forecast accuracy of a multivariate series $\mathbf{f}_t \in \mathbb{R}^n$ by creating linear components from it and forecasting them too



Component weights matrix chosen by forecaster:

For example, do PCA on $(\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_{\mathrm{T}})$ and use the loadings in Φ

Why could this improve forecast accuracy?

While $\boldsymbol{c}_t$ has no new underlying data, the forecasts $\hat{\mathbf{c}}_{\mathrm{T}+h|\mathrm{T}}$ may capture information not in $\hat{\mathbf{f}}_{\mathrm{T}+h|\mathrm{T}}$

Research Question: *To what extent* can forecasts based on constructed linear components improve the accuracy of hierarchical forecasts when incorporated into the Information Combination framework?

Forecasting with Linear Augmented Information Combination

1. Create p linear component series from $\mathfrak{T}_T$ using PCA and forecast them → 2. Apply IComb framework on set of augmented predictors, to get FLAIC coefficient matrix → 3. Use obtained coefficients to reconcile base forecast

- We use all *four* PCA specifications. Let $\delta_C, \delta_S \in \{0,1\}$ binary indicators for centring and scaling
- Define

$$\mathbf{y}_t^{(\delta_C,\delta_S)} = \mathbf{D}_{(\delta_C)}^{-\delta_S}(\mathbf{y}_t - \delta_C\bar{\mathbf{y}})$$

  where: $\mathbf{D}_{(\delta_C)}$ is diagonal matrix of scaling factors

- Apply PCA to $\mathbf{Y}_{\mathrm{T}}^{(\delta_C,\delta_S)}$ construct $p \leq \min(T-h-\delta_C, n_b)$ components as:

$$\mathbf{c}_t^{(\delta_C,\delta_S)} = \mathbf{\Phi}_p^{(\delta_C,\delta_S)}\left(\mathbf{D}_{(\delta_C)}^{-\delta_S}(\mathbf{y}_t - \delta_C\bar{\mathbf{y}})\right)$$

PC scores *PC Loadings*

- Forecast components, $\hat{\mathbf{c}}_{t+h|t}$

Forecasting with Linear Augmented Information Combination …*(cont'd)*

- Create an augmented predictor set, $\mathbf{Z}$, for responses, $\mathbf{Y}$

$$\hat{\mathbf{z}}_{t+h|t} = \begin{bmatrix} \hat{\mathbf{c}}_{t+h|t} \\ \hat{\mathbf{y}}_{t+h|t} \end{bmatrix} \qquad \text{stacked:} \qquad \mathbf{Z} = \begin{bmatrix} \hat{\mathbf{z}}'_{1+h|1} \\ \hat{\mathbf{z}}'_{2+h|2} \\ \vdots \\ \mathbf{z}'_{T|T-h} \end{bmatrix}, \quad \text{with} \quad \mathbf{Y} = \begin{bmatrix} \mathbf{y}'_{1+h} \\ \mathbf{y}'_{2+h} \\ \vdots \\ \mathbf{y}'_{T} \end{bmatrix}$$

- Apply IComb framework, replacing $\mathbf{X}$ with $\mathbf{Z}$ only: $\mathbf{Y} = \mathbf{Z}\,\mathbf{B} + \mathbf{E}$
- Reconciled $\tilde{\mathbf{y}}^{FLAIC}_{\mathrm{T}+h|\mathrm{T}} = \left(\widehat{\mathbf{B}}^{\mathrm{FLAIC}}\right)' \hat{\mathbf{z}}_{\mathrm{T}+h|\mathrm{T}}$ is coherent

- Challenges motivate regularisation
  1. Additional p coefficients to be estimated in each of the n regressions
  2. Multicollinearity: augmented predictor set may have highly correlated or redundant signals
     - ✓ FLAIC-S: Ridge (shrinkage) ↔ dense models
     - ✓ FLAIC-L: Multivariate lasso ↔ sparse models

- Potential improvement in forecast accuracy:
  1. The forecasts, $\hat{\mathbf{c}}_{t+h|t}$ may capture information not in $\hat{\mathbf{y}}_{t+h|t}$
  2. Components $\mathbf{c}_t$ may exhibit more forecastable dynamics

2.0 Empirical Investigation

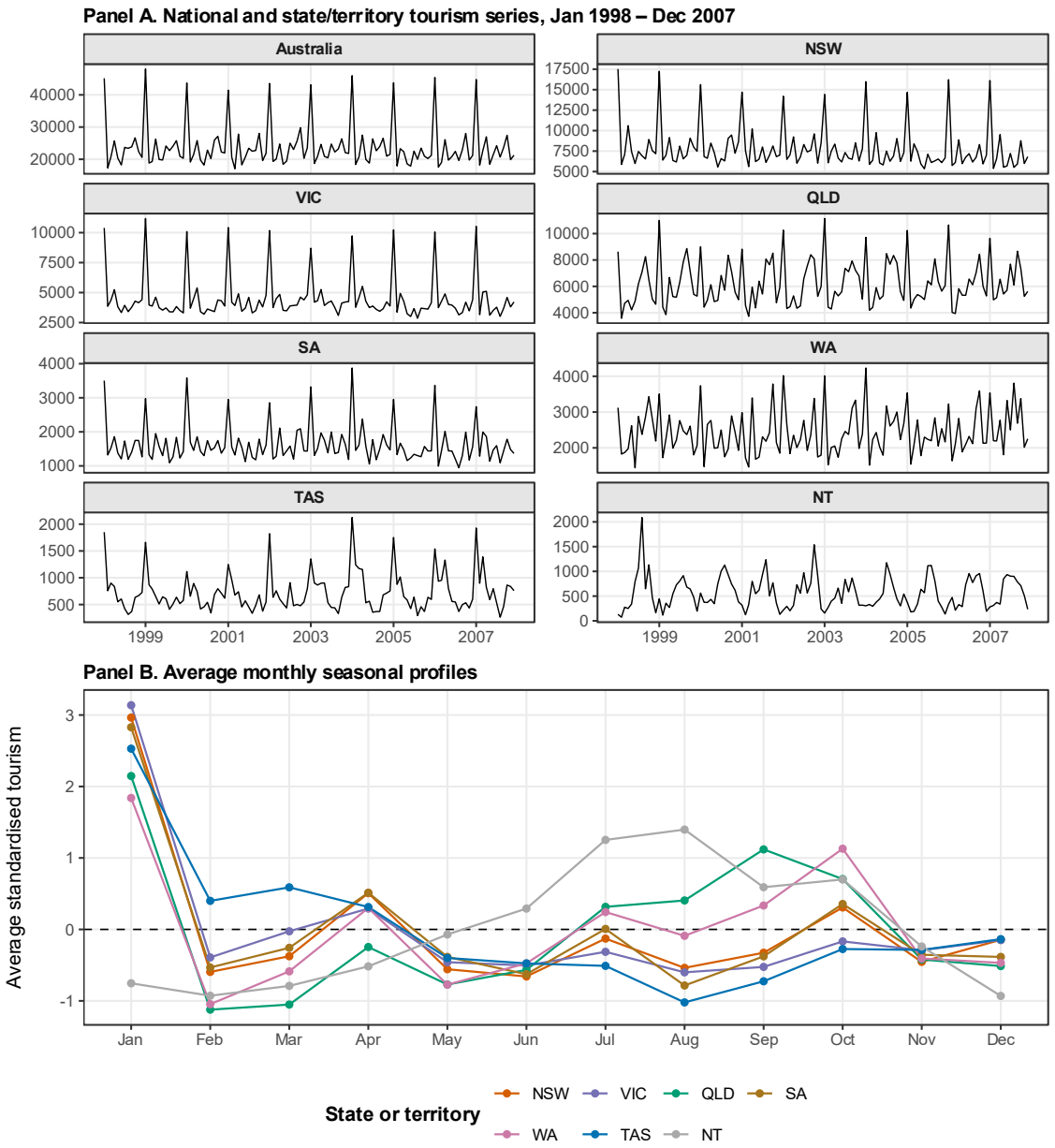


Figure. *Australian domestic tourism series and seasonal profiles*

- Australian domestic tourism visitor nights (Tourism Research Australia, 2026)

- 4 levels: national, states & territories, zones, regions
- 112 series, $n_b = 77$ regions
- 264 months: Jan 1998 – Dec 2019

- Evidence of seasonality and state-level differences

- Note: forecasting this data (and components):
  - 120 month rolling windows (index by origins)
  - Error-Trend-Seasonality (ETS) models
  - $h = 1,2, \dots, 12$ month ahead out-of-sample

    (short-term: $h = 1 - 6$ & long-term $h = 1 - 12$)
  - $144 - h$ available forecast origins

Components created for all four PCA (centre, scale) specifications

PCA specification	PC	Share (%)	Seasonal R^2 (%)
Centred, unscaled	PC1	93.1 [92.6, 93.5]	93.8 [74.7, 96.0]
	PC2	3.4 [3.1, 3.7]	79.7 [77.2, 83.2]
	PC3	0.8 [0.5, 1.4]	24.9 [8.6, 34.0]
Centred, scaled	PC1	38.7 [38.0, 40.0]	94.4 [78.2, 95.8]
	PC2	13.3 [12.2, 14.9]	86.0 [72.9, 89.4]
	PC3	3.2 [2.8, 3.9]	39.4 [15.3, 58.1]
Uncentred, unscaled	PC1	99.3 [99.3, 99.4]	93.5 [74.0, 95.8]
	PC2	0.4 [0.3, 0.4]	91.1 [89.1, 93.3]
	PC3	0.1 [0.1, 0.1]	42.9 [6.7, 52.3]
Uncentred, scaled	PC1	81.6 [81.2, 83.8]	90.2 [69.3, 92.6]
	PC2	5.6 [5.1, 5.9]	94.4 [92.9, 94.9]
	PC3	1.0 [0.9, 1.1]	68.0 [64.0, 71.0]

Table: Median [2.5th, 97.5th percentiles] across rolling windows. Share denotes percentage of total sum of squares. Seasonal R^2 is from month-of-year regressions.

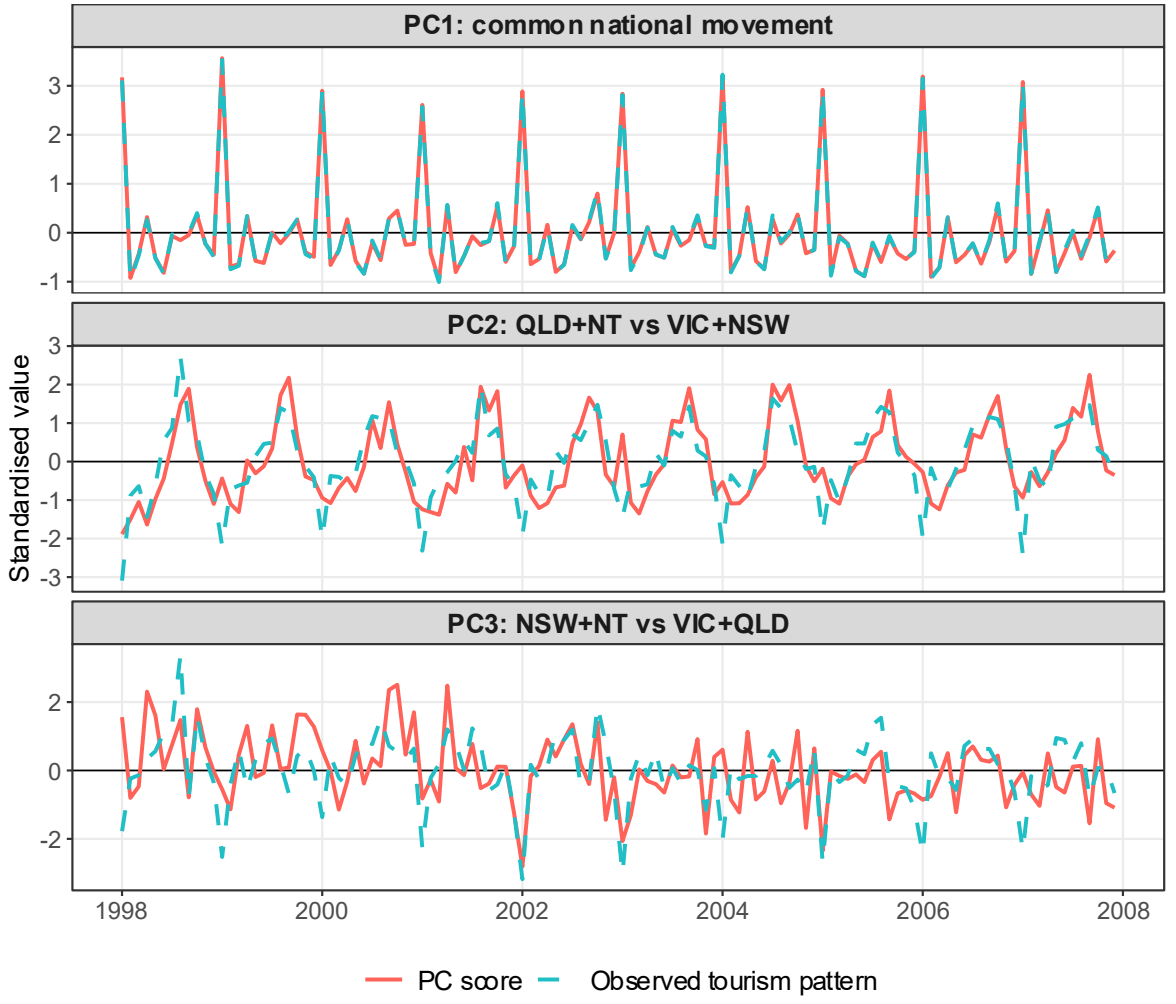


Figure: Illustrative patterns from centred, unscaled PCA using the initial rolling window. Series are standardised for visual comparison.

Each rolling training window:

- Conventional method reconciliation: BU, OLS, WLS_v, WLS_s, MinT_{shr}

- IComb
  - Ridge (IComb-S) and lasso (IComb-L) regularisation estimation
  - Implementation:
    - $\mathbf{X}$ and $\mathbf{Y}$ are standardised: $\mathbf{X}^* = \mathbf{X}\,\boldsymbol{\Sigma}_x^{-0.5}$, $\mathbf{Y}^* = \mathbf{Y}\,\boldsymbol{\Sigma}_y^{-0.5}$, $\hat{\mathbf{B}} = \boldsymbol{\Sigma}_x^{-0.5}\,\hat{\mathbf{B}}^*\,\boldsymbol{\Sigma}_y^{0.5}$
    - Intercept included and not regularised
    - Tuning parameter (λ) selected from a grid, based on expanding window cross-validation minimising mean squared forecast error of validation set

- FLAIC
  - 64 variants:

$$4 \text{ PCA specifications} \quad \times \quad p \in \{\underbrace{1,2,3,5,}_{\text{low}} \ \underbrace{10,20,}_{\text{moderate}} \ \underbrace{40,77}_{\text{high}}\} \quad \times \quad \{\text{FLAIC-S, FLAIC-L}\}$$

  - Component augmented predictor set in IComb (replace $\mathbf{X}$ with $\mathbf{Z}$) with same implementation settings

2.4.1 PRIAL

- Forecast performance measure: percentage relative improvement in average loss (PRIAL) (Nguyen et al., 2025):

- Let m index method. MSFE computed over all series (Average) and rolling forecast origins.*

$$\text{PRIAL}_{m,h} = \frac{\text{MSFE}_{m,h} - \text{MSFE}_{\text{base},h}}{\text{MSFE}_{\text{base},h}} \times 100$$

- Key findings:
  - Best *overall*: FLAIC-L
    - Earlier horizons ($h = 1$ to 6): centred, unscaled ($p = 3$ or 5)
    - Later horizons ($h = 7$ to 12): centred, scaled ($p = 10$ or 20)
  - Best FLAIC-S has similar PCA specification and p as FLAIC-L
  - Does not necessarily indicate component augmentation more effective under lasso

PRIAL for long-term period ($h = 1 - 12$):

- *Best FLAIC-L:* centred, unscaled, p=5
- *Best FLAIC-S:* centred, scaled, p=10

Regularisation/Method	IComb	FLAIC	Difference
Ridge	-4.65%	-5.82%	1.17 pp
Lasso	-6.25%	-6.67%	0.42 pp
Difference	1.6 pp	0.85 pp	

** Note: We also computed MSFE using scaled errors; results in Appendix slides.*

2.4.2 Δ PRIAL

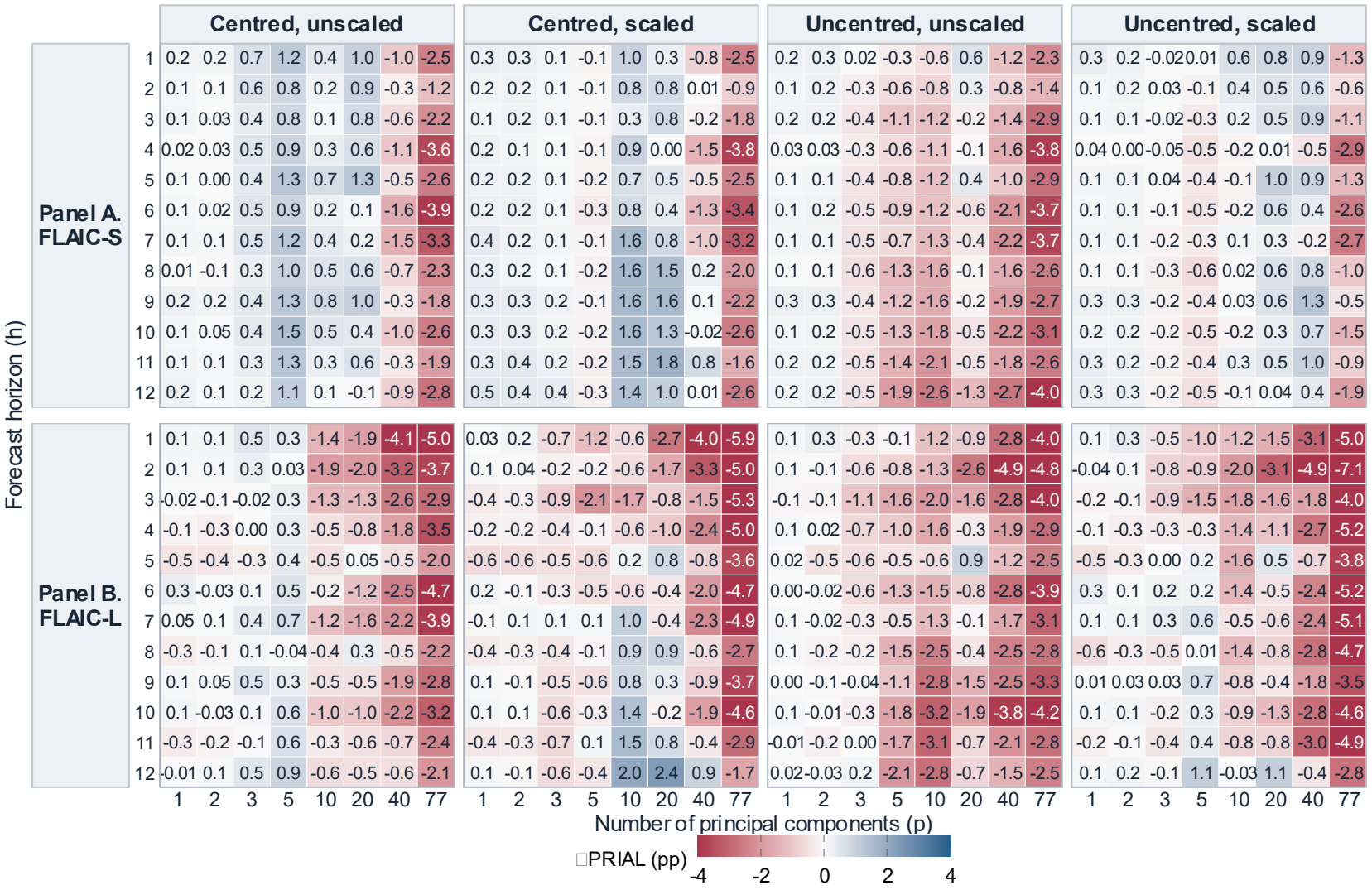


Figure: *ΔPRIAL*: FLAIC relative to matched IComb: Average across 112 hierarchy series. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L.

- When does component augmentation improve upon IComb?

$$\Delta\text{PRIAL}_h = \text{PRIAL}_{\text{IComb},h} - \text{PRIAL}_{\text{FLAIC},h}$$

$$= \frac{\text{MSFE}_{\text{IComb},h} - \text{MSFE}_{\text{FLAIC},h}}{\text{MSFE}_{\text{base},h}} \times 100$$

- $\Delta\text{PRIAL} > 0 \implies$ FLAIC better

- Findings:
  1. Effects of component augmentation depends on p
  2. Component augmentation generally more effective for ridge than lasso*
  3. Direction of relative performance often maintained across h

* Generally, FLAIC-L retains many higher order component forecasts, without reducing the number of retained hierarchy base forecasts retained by IComb-L. *Diagnostics available in Appendix slides.*

2.4.2 Δ PRIAL *(decomposed)*

- PRIAL depends on MSFE.

$$\text{MSFE}_{m,h} = \frac{1}{N_h n} \sum_{\tau \in \mathcal{O}_h} \sum_{i=1}^{n} \left(\tilde{e}_{\tau,h,i}^{(m)} \right)^2 = \frac{1}{N_h} \sum_{\tau \in \mathcal{O}_h} \ell_{\tau,h}^{(m)}$$

- Therefore:

$$\Delta\text{PRIAL}_h = \frac{\text{MSFE}_{\text{IComb},h} - \text{MSFE}_{\text{FLAIC},h}}{\text{MSFE}_{\text{base},h}} \times 100$$

$$= \frac{1}{N_h} \frac{\sum_{\tau \in \mathcal{O}_h} 100 \left(\ell_{\tau,h}^{(\text{IComb})} - \ell_{\tau,h}^{(\text{FLAIC})} \right)}{\text{MSFE}_{\text{base},h}}$$

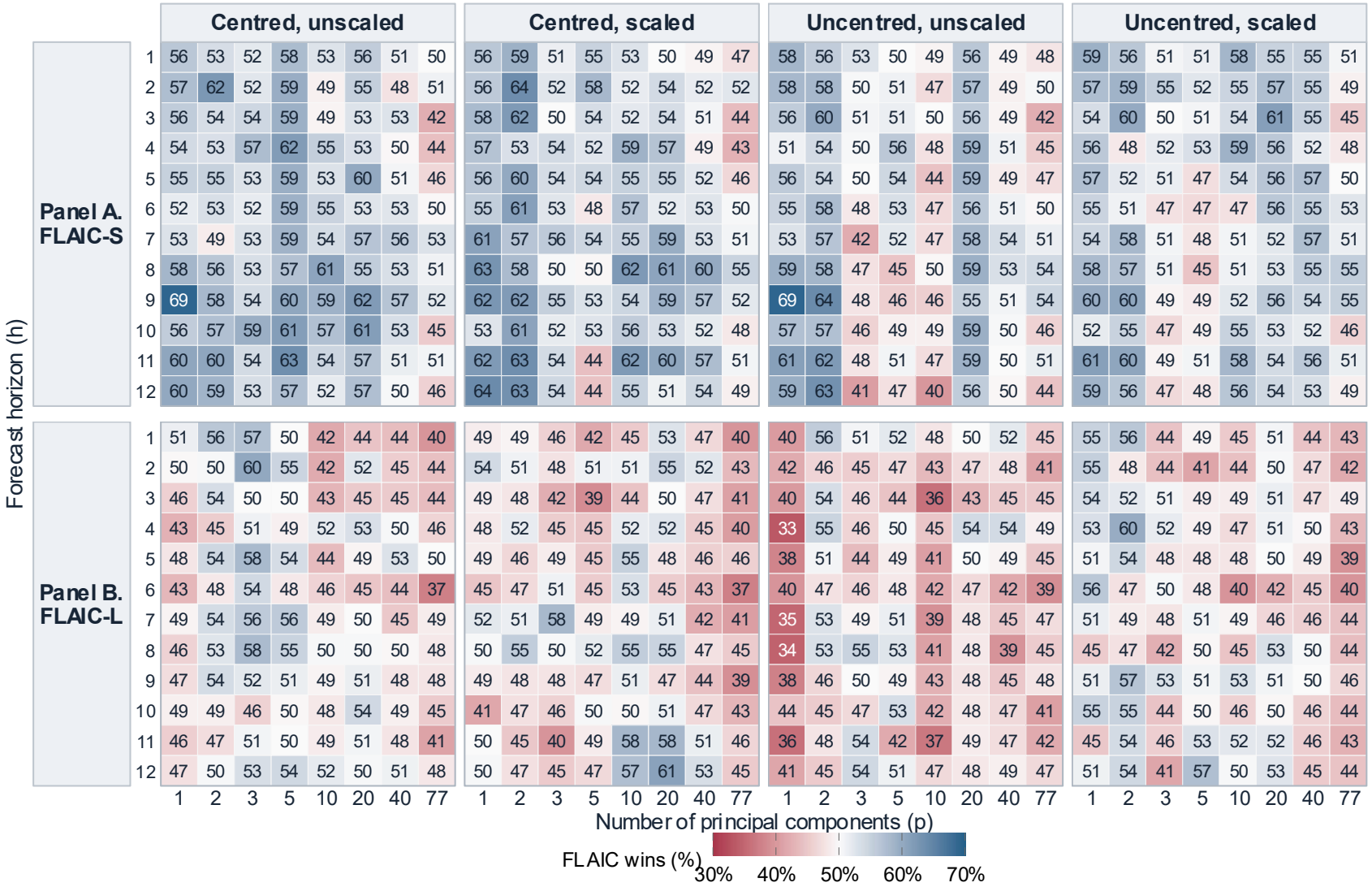
(2.4.3)
Origin-level loss difference > 0
$\Longrightarrow$ FLAIC better for that origin

Statistic: % origins FLAIC better >50% $\Longrightarrow$ FLAIC better more origins

(2.4.4)
$q_{\tau,h}$ = origin-level contributions to ΔPRIAL_h (magnitudes across origins)

- Alternatively, average squared errors across rolling window origins and compare for series by levels *(2.4.5)*
- Statistically significant forecast accuracy gains? *(2.4.6)*

2.4.3 *Rolling forecast origin wins (FLAIC v IComb)*



- Agrees with ΔPRIAL_h findings
- Cell-wise comparison for both heatmaps
  - ΔPRIAL and FLAIC wins match 74.2% (FLAIC-S) and 66.9% (FLAIC-L)
  - Most differences arise when ΔPRIAL < 0, but FLAIC wins > 50%

Figure: Percentage of rolling forecast origins at which FLAIC has a lower average squared-error loss than matched IComb, averaged across the 112 hierarchy series. FLAIC-S and FLAIC-L are compared with IComb-S and IComb-L, respectively. Percentages are rounded to integers.

2.4.4 Origin–level contributions to Δ PRIAL

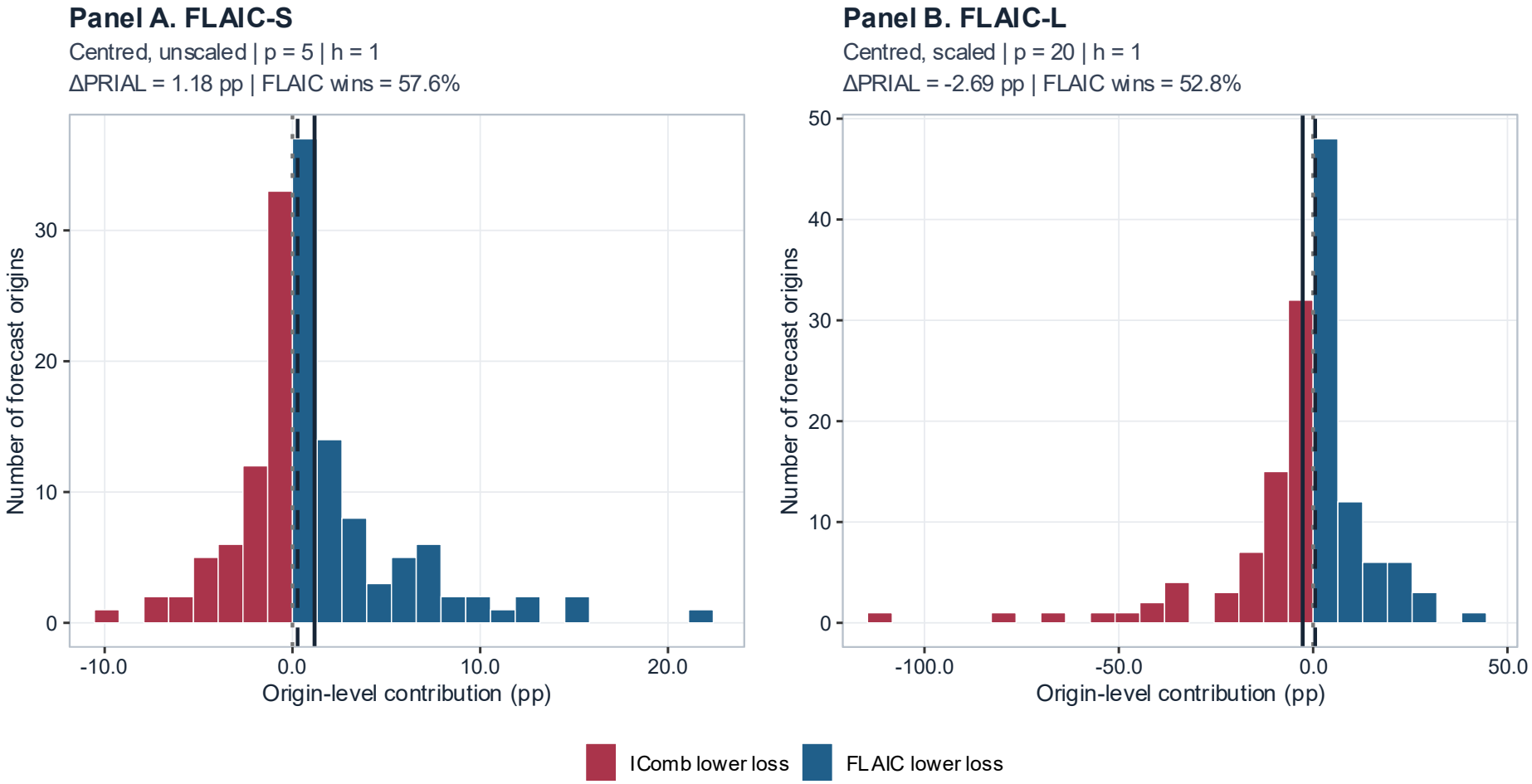
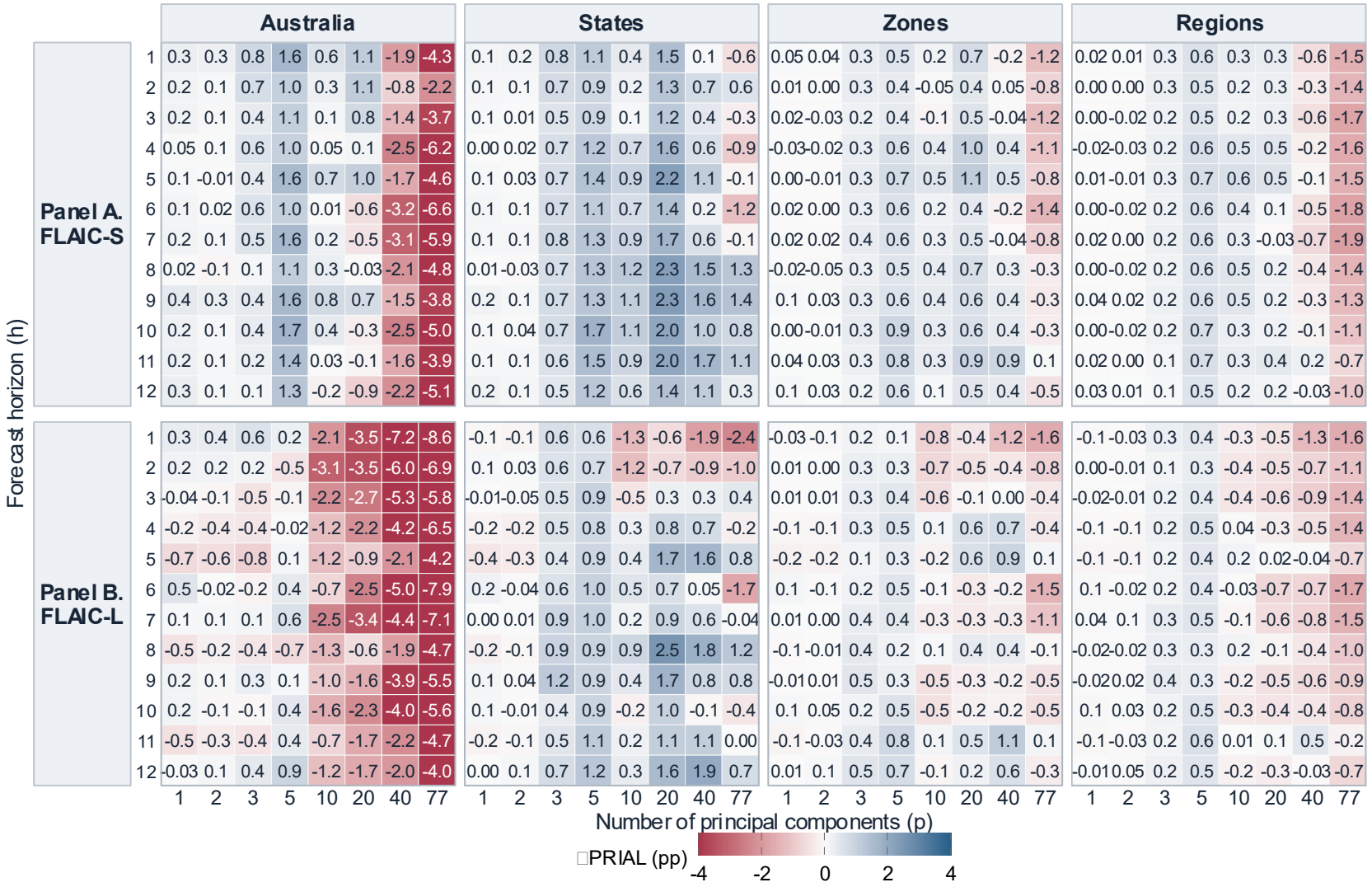


Figure: Distributions of origin-level contributions to Δ *PRIAL*, after squared error loss is averaged across the 112 hierarchy series. Positive values favour FLAIC. Average of origin-level contributions equals Δ *PRIAL*.

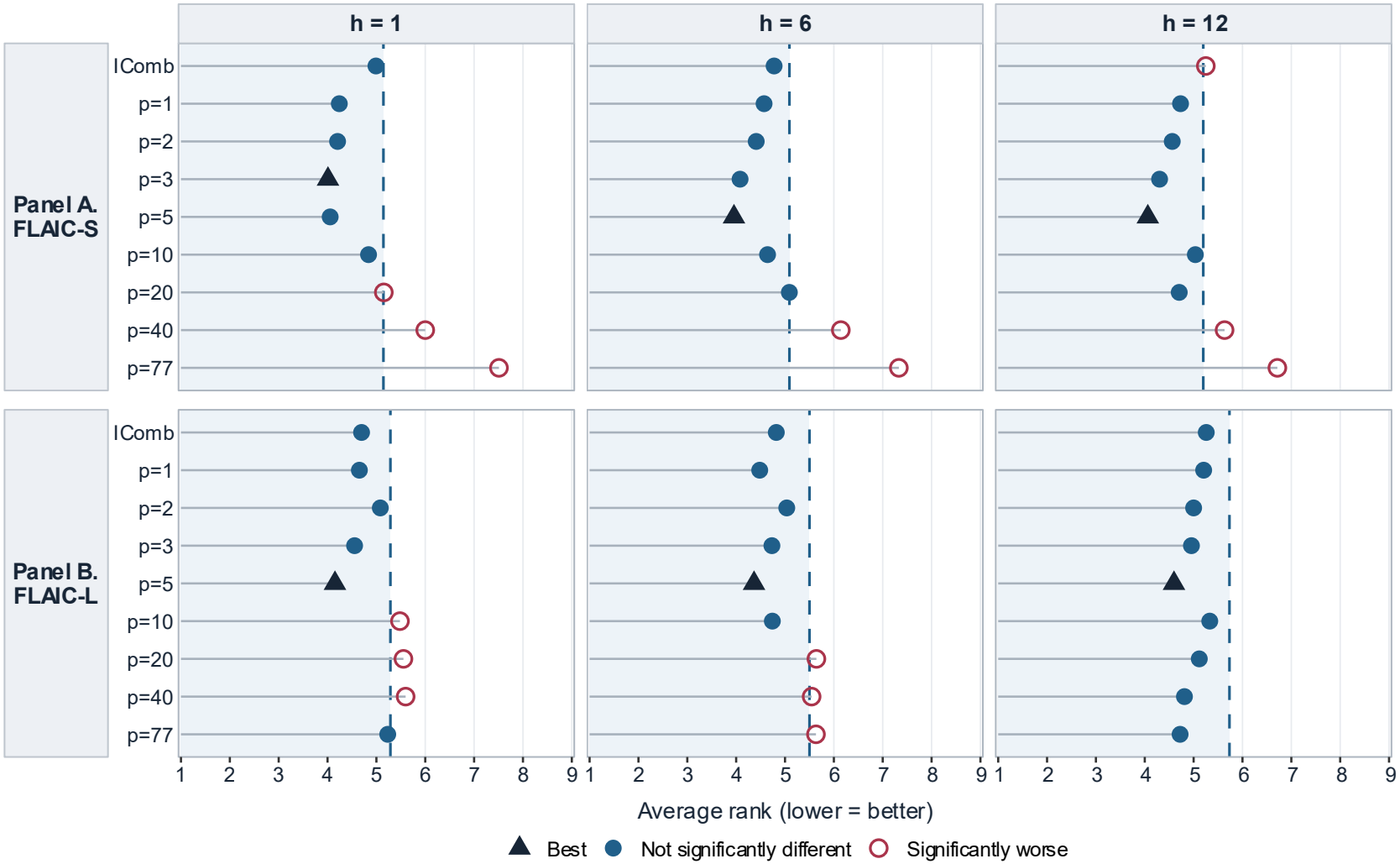
2.4.5 Δ PRIAL and PRIAL by hierarchy level



- Heterogeneous effects across hierarchy levels
  - ✓ States tend to benefit more
- Higher dimensional component augmentation mostly deteriorates relative forecast performance
- Relative to base forecast (PRIAL):
  - Best PCA specification, p, ridge/lasso vary by level

Figure: *ΔPRIAL*: FLAIC relative to matched IComb (using centred, unscaled PCA), computed for each hierarchy level. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L, and positive values favour FLAIC over matched IComb.

2.4.6 Statistical significance visualised via MCB



- Rank each series across methods to get a distribution of ranks for each method. Each method is summarised by its average rank across series.
  (Friedman 1937, 1939; Nemenyi 1963; Hollander et al. 2013; Koning et al. 2005)

Considering all 4 PCA specifications:

- Best-ranked method generally has low component augmentation
- Limited evidence that improvement in forecast accuracy under FLAIC is statistically significant (especially lasso)
- High-dimensional component augmentation is typically statistically significantly worse (especially ridge)

Figure: Multiple comparison with best (MCB): comparison of average ranks against best-rank method. Uses average MSFE ranks across 112 hierarchy series at $h = 1,6,12$ for centred, unscaled PCA specification.

2.4.7 *Hierarchy-adapted FLAP benchmark*

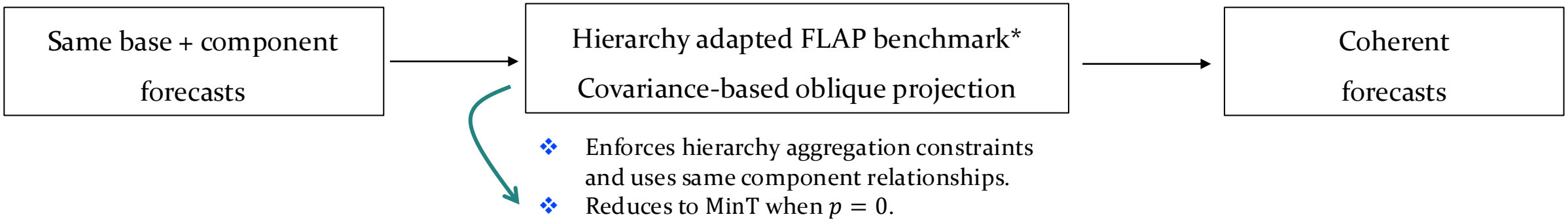


Table: component augmentation across reconciliation methods (selected)
Centred, Scaled PCA | PRIAL (%) | $h = 1$

Method	**No PCs***	$p = 5$	$p = 10$	$p = 20$	$p = 40$	$p = 77$
$\text{MinT}_{shr} \rightarrow$ FLAP	5.00	1.16	0.35	-0.29	-0.55	-0.08
IComb − S → FLAIC − S	−4.98	−4.85	−5.97	−5.26	−4.15	−2.45
IComb − L → FLAIC − L	−7.49	−6.25	−6.87	−4.81	−3.52	−1.56

Notes: *No PCs means $p = 0$ and identifies the corresponding MinT or IComb benchmark.
Lower PRIAL = better forecast accuracy.

Findings:

- Component augmentation through FLAP consistently has better accuracy than MinT_{shr}
- FLAIC achieves better absolute accuracy than FLAP in the reported matched comparisons.
- The benefits of component augmentation depend on the reconciliation method.

** More details in appendix slides*

Concluding remarks:

- Research question: *To what extent* can forecasts based on constructed linear components improve the accuracy of hierarchical forecasts when incorporated into the Information Combination framework?

- Answer: It can, under selected specifications — but improvements in forecast accuracy are not consistently statistically significant.

- Our contribution is mainly methodological:
  - Propose FLAIC: incorporates separately forecast principal components into IComb framework, while maintaining coherence and without requiring additional underlying data.
  - Develop a hierarchy-adapted FLAP benchmark, incorporating both component and aggregation constraints.

- Implications – established via empirical investigation
  - Low/moderate augmentation can improve accuracy of FLAIC relative to IComb, particularly under ridge; very high augmentation generally deteriorates FLAIC's relative forecast performance.
  - Benefits of component augmentation depend on PCA specification, regularisation, and reconciliation method.

Limitations and future research

Limitations of our research:

- One empirical dataset (Australian domestic tourism) and ETS forecasts
- Unclear when component forecasts offer genuinely useful complementary information
- Predefined PCA specifications, component augmentation dimensions and one IComb implementation.

Identifying the conditions under which component augmentation provides a reliable improvement in forecast accuracy relative to IComb is an important direction for future research:

- Monte Carlo simulations varying hierarchy structure, base forecasting models, common signals, forecast error correlation and dimensionality
- Use different IComb framework variants (example, without standardisation)
- If best FLAIC variant is an empirical question, then may need data-driven selection for PCA specification and augmentation dimension

Appendix

Lasso diagnostics

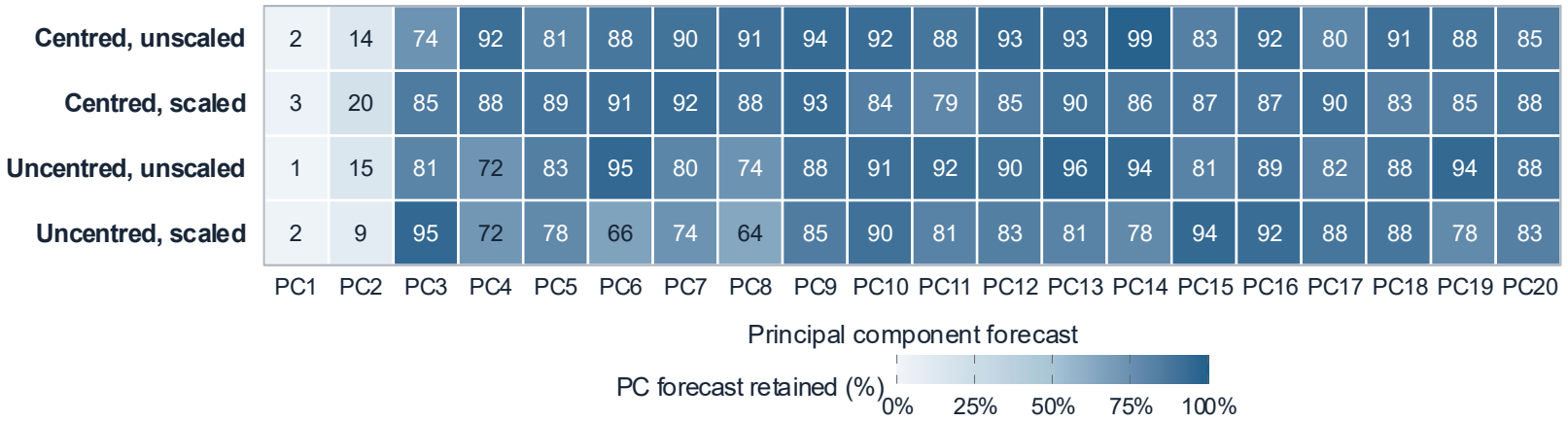


Figure: Cells report the percentage of 144 rolling forecast origins at which each PC forecast is retained by FLAIC-L. Retention means at least one non-zero coefficient across the 112 response equations.

Findings for component forecast retention:

- PC1 and PC2 rarely retained while higher order PCs frequently kept
- Component forecast retention is not accompanied by a comparable reduction in hierarchy base forecasts.
- Retention does not consistently translate into accuracy gains.

Component forecasts retained (median) by FLAIC-L: (number, %)

16 / 20 (80%)

Δ hierarchy base forecasts retained relative to IComb-L median:

– 2

Example: At $p = 20$, centred and unscaled PCA

*Note: IComb-L retained a median of 52 of 112 hierarchy base forecasts

Robustness: Using scaled errors

- Series at hierarchy levels differ substantially in scale.
- Compute mean squared *scaled* forecast error (MSSFE):

$$\text{MSSFE}_{m,h} = \frac{1}{N_h n} \sum_{\tau \in \mathcal{O}_h} \sum_{i=1}^{n} \frac{\left(\tilde{e}_{\tau,h,i}^{(m)}\right)^2}{d_{\tau,i}}$$

using two scaling denominators from 120-month training window:

1. Seasonal naïve forecast: $d_{\tau,i}^{(SN)} = \frac{1}{108} \sum_{t=\tau-107}^{\tau} \left(y_{t,i} - y_{t-12,i}\right)^2$
2. Variance standardised: $d_{\tau,i}^{(V)} = \frac{1}{120} \sum_{t=\tau-119}^{\tau} \left(y_{t,i} - \bar{y}_{\tau,i}\right)^2$

Finding	**Raw error measure**	**Both scaled-errors measures**
Low/moderate augmentation can help	Yes	Yes (but ΔPRIAL smaller magnitude)
High augmentation generally hurts	Yes	Yes
Best-performing FLAIC variant	Lasso	Ridge

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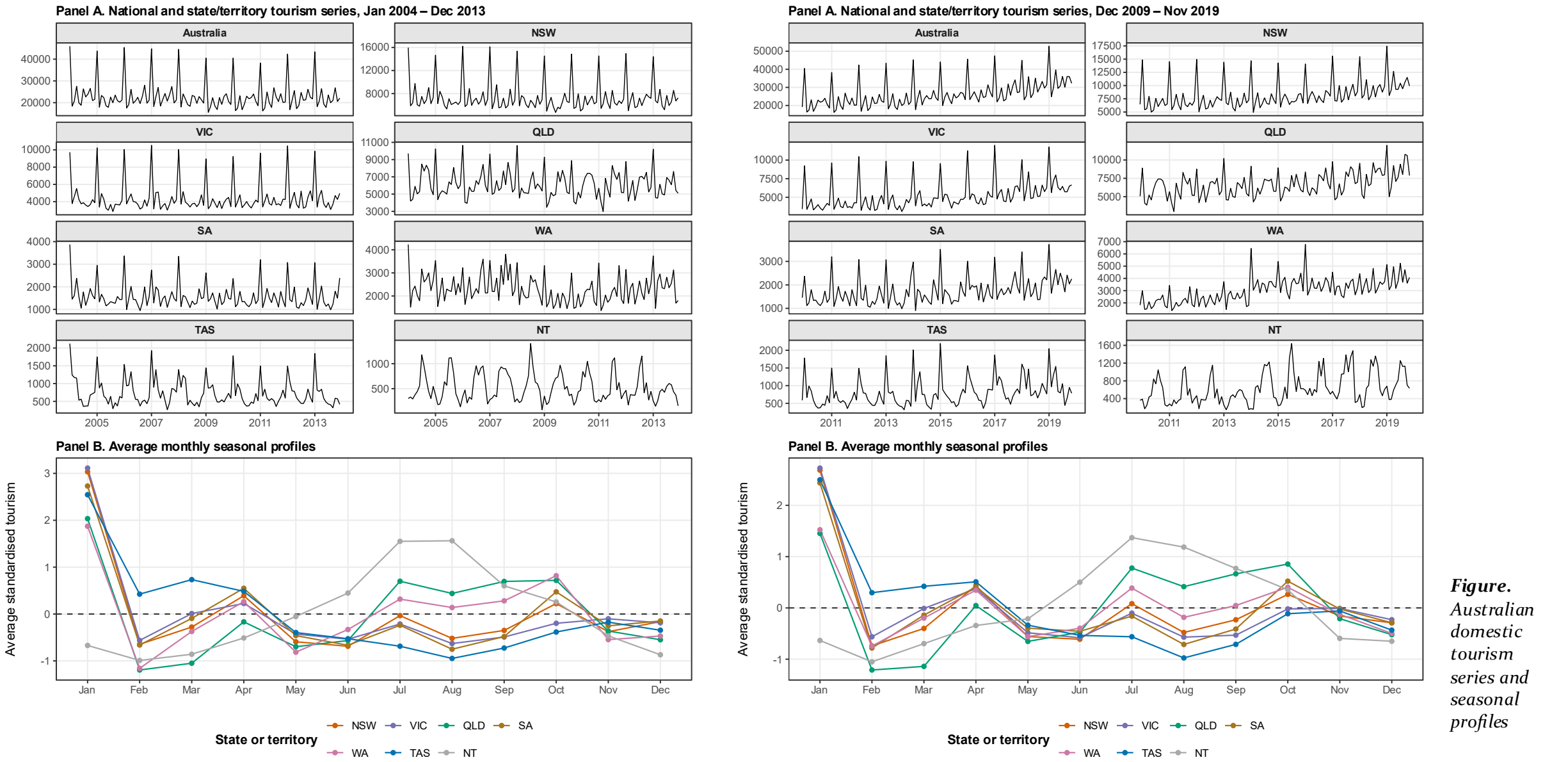
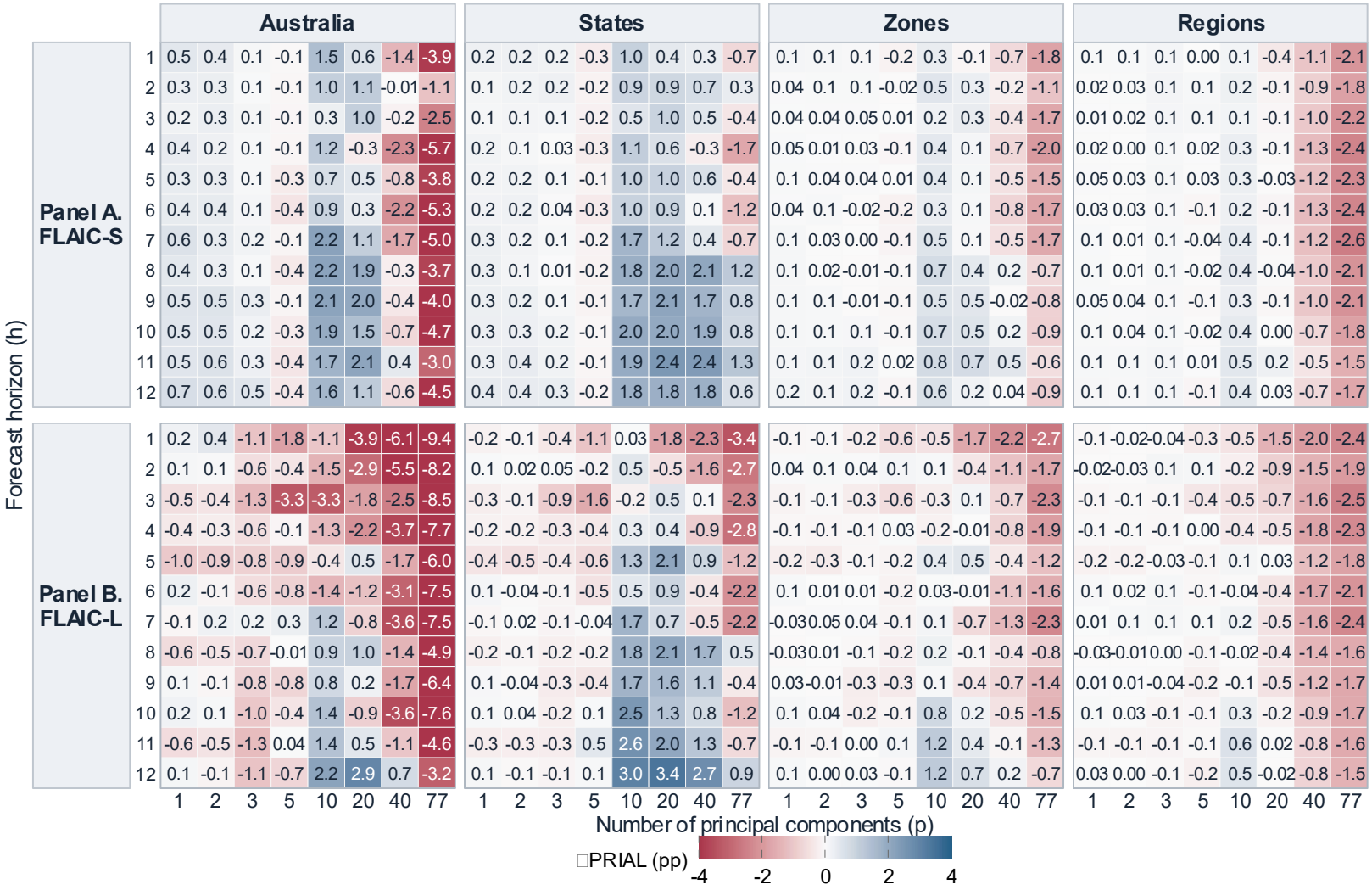


Figure. *Australian domestic tourism series and seasonal profiles*

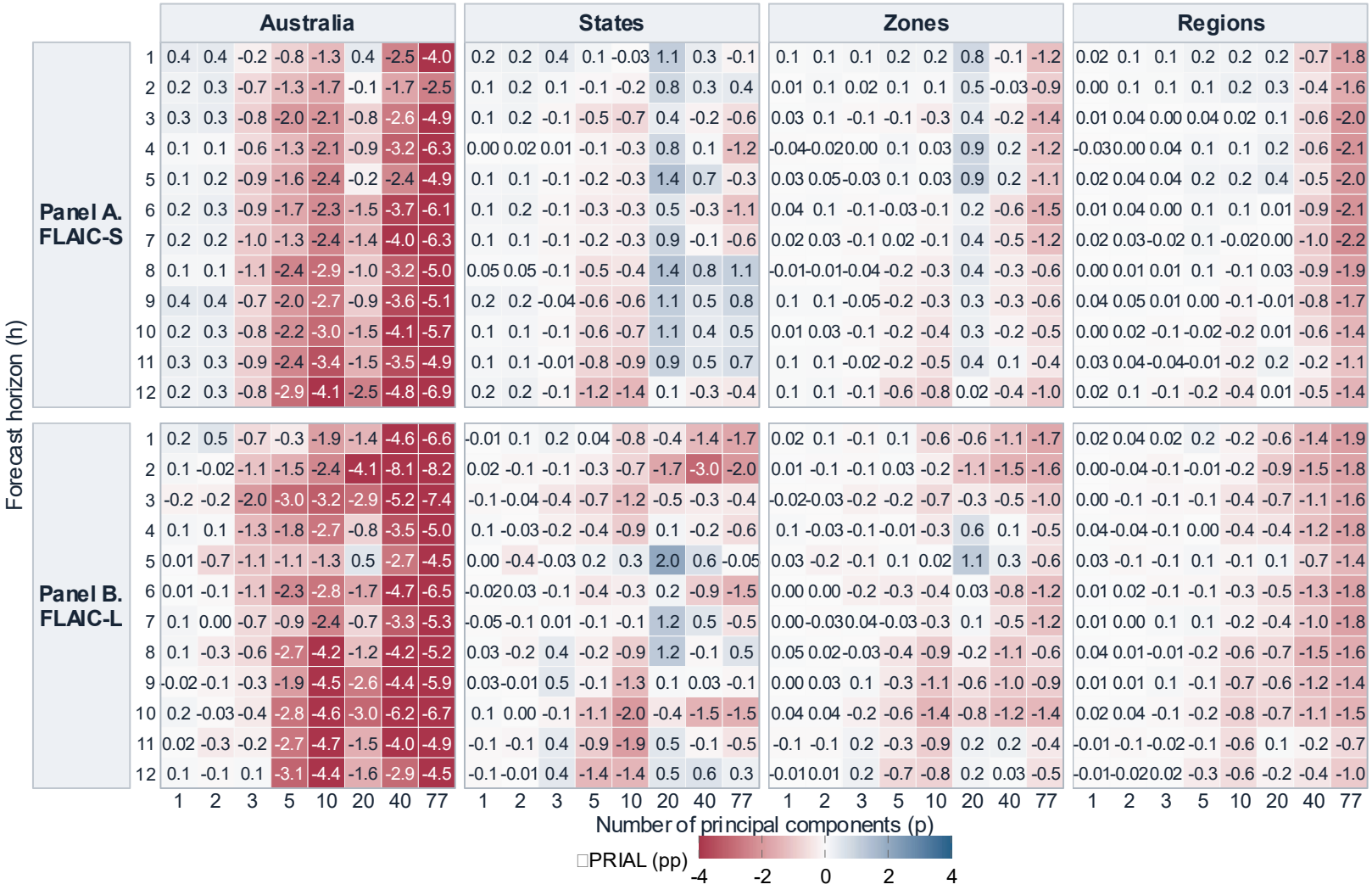
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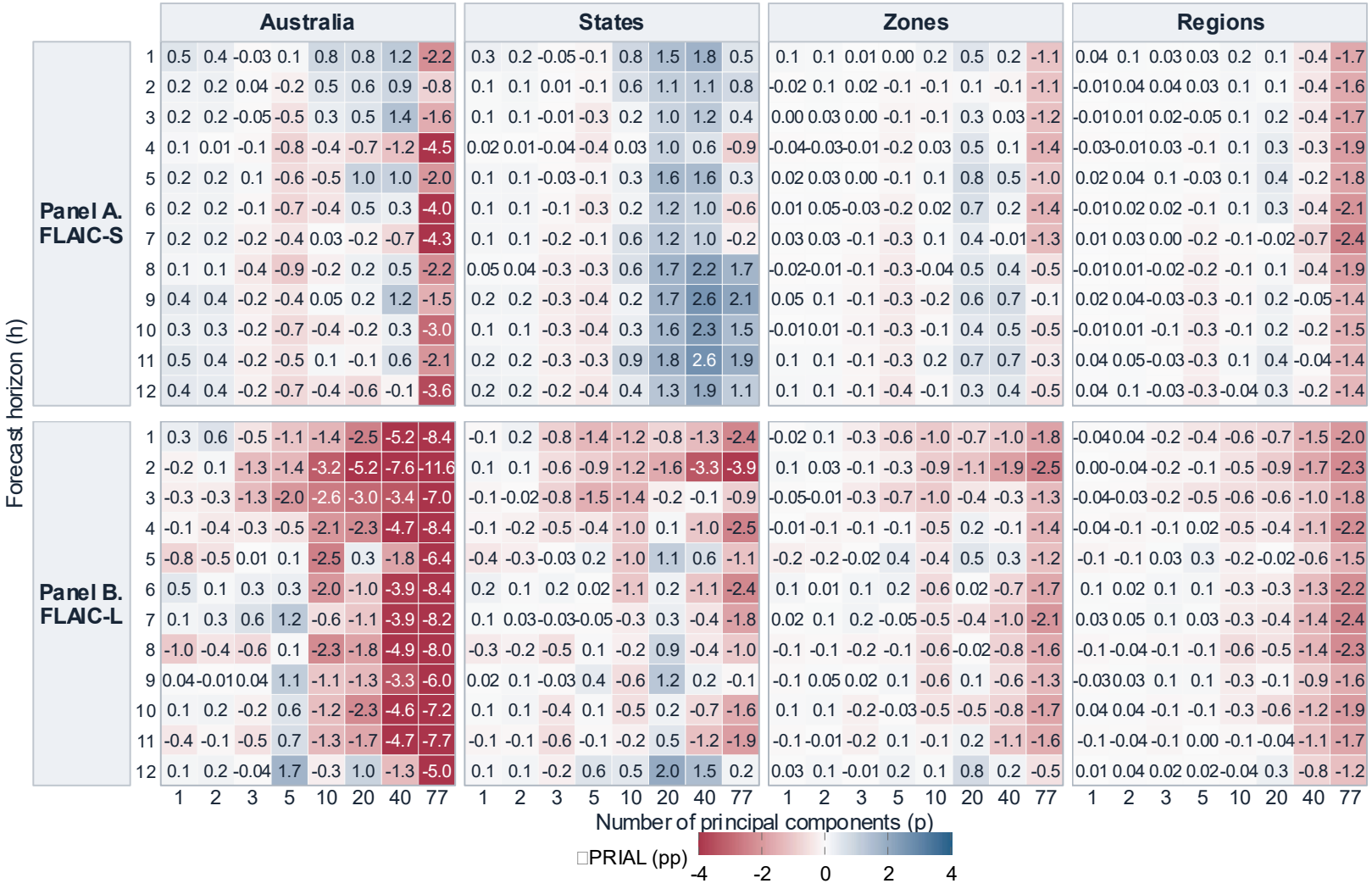
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  - Best PCA specification, p, ridge/lasso vary by level

Figure: *ΔPRIAL*: FLAIC relative to matched IComb (**using uncentred, unscaled PCA**), computed for each hierarchy level. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L, and positive values favour FLAIC over matched IComb.

2.4.4 Δ PRIAL and PRIAL by hierarchy level



- Heterogeneous effects across hierarchy levels
  - ✓ States tend to benefit more
- Higher dimensional component augmentation mostly deteriorates relative forecast performance
- Relative to base forecast (PRIAL):
  - Best PCA specification, p, ridge/lasso vary by level

Figure: *ΔPRIAL*: FLAIC relative to matched IComb (**using uncentred, scaled PCA**), computed for each hierarchy level. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L, and positive values favour FLAIC over matched IComb.

2.4.5 Statistical significance visualised via MCB

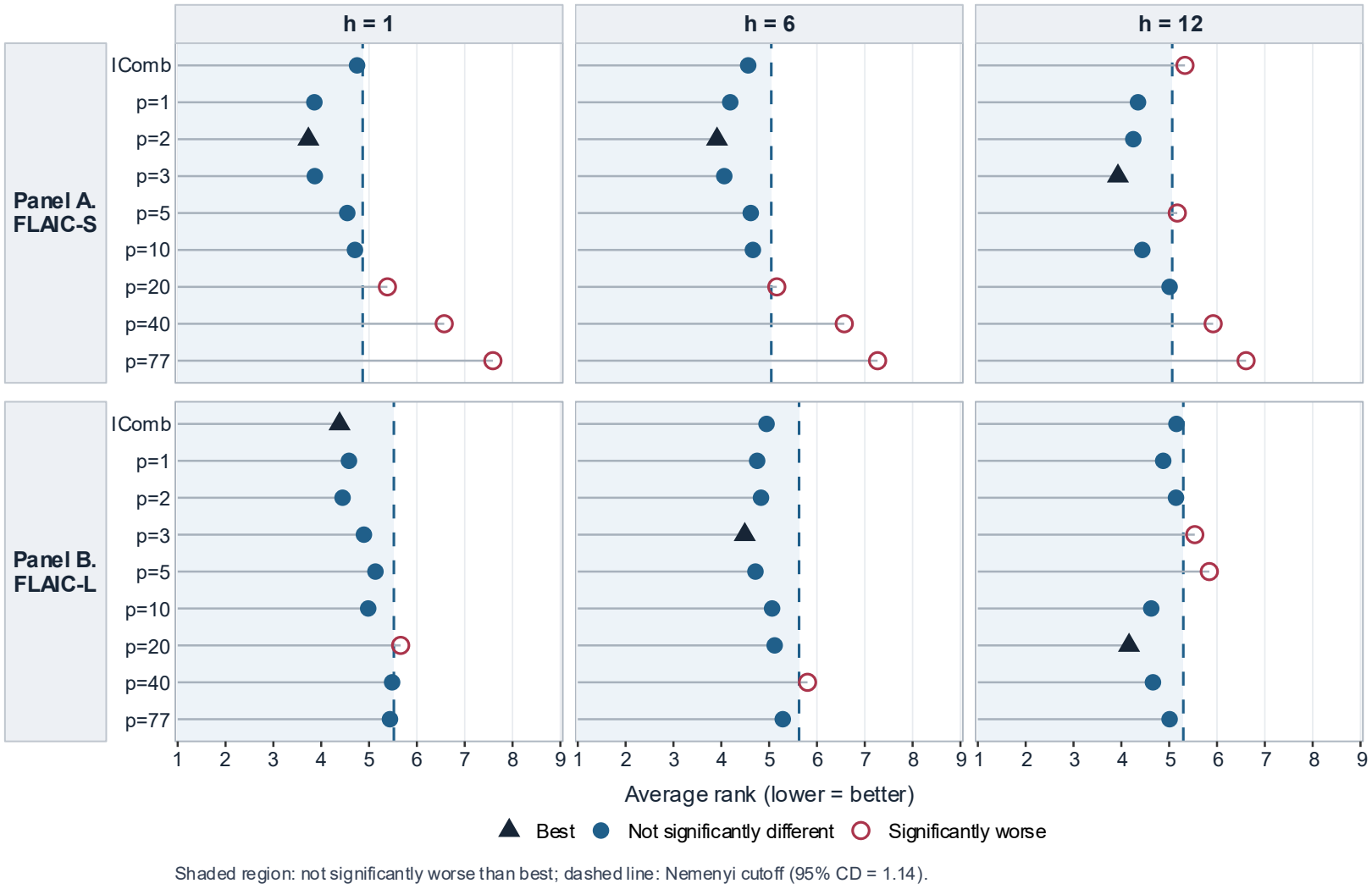


Figure: Multiple comparison with best (MCB): comparison of average ranks against best-rank method. Uses average MSFE ranks across 112 hierarchy series at $h = 1,6,12$ for **centred, scaled PCA specification**.

$$\text{MSFE}_{m,h,i} = \frac{1}{N_h} \sum_{\tau \in \mathcal{O}_h} \left(\tilde{e}_{\tau,h,i}^{(m)} \right)^2$$

- Rank each series across methods to get a distribution of ranks for each method. Each method is summarised by its average rank across series.
  (Friedman 1937, 1939; Nemenyi 1963; Hollander et al. 2013; Koning et al. 2005)

Considering other 3 PCA specs too:

- Best-ranked method tends to be low augmentation
- Limited evidence that improvement in forecast accuracy under FLAIC is statistically significant (especially lasso)
- High-dimensional component augmentation typically statistically significantly worse (especially ridge)

2.4.5 Statistical significance visualised via MCB

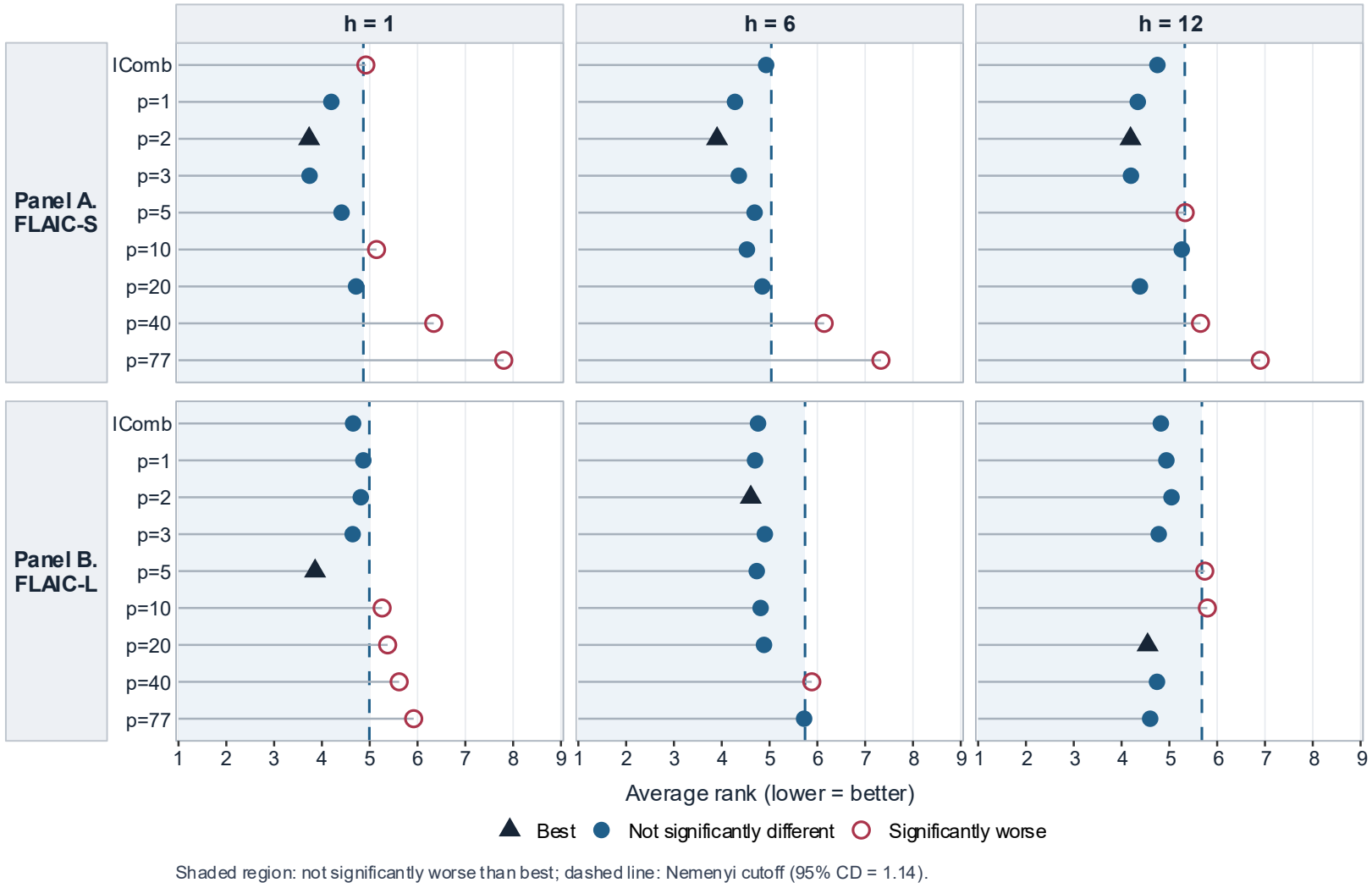


Figure: Multiple comparison with best (MCB): comparison of average ranks against best-rank method. Uses average MSFE ranks across 112 hierarchy series at $h = 1,6,12$ for **uncentred, unscaled PCA specification**.

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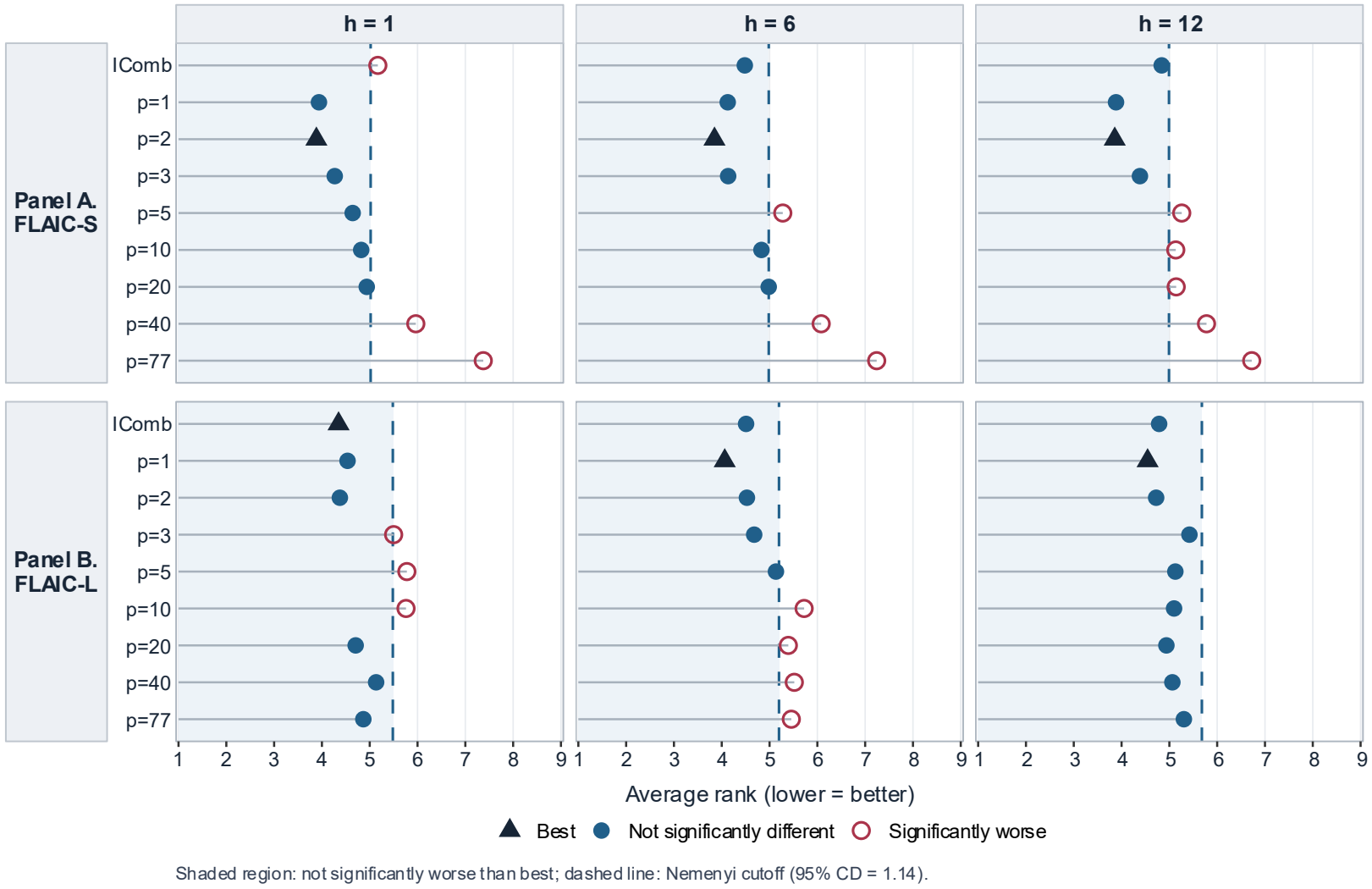


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2.4.7 *Hierarchy-adapted FLAP benchmark*

Same base + component forecasts → Hierarchy adapted FLAP benchmark: Covariance-based oblique projection → Coherent forecasts

- Enforces hierarchy aggregation constraints and uses same component relationships.
- Reduces to MinT when $p = 0$.

(1) Write our component augmented vector as:

$$\begin{bmatrix} \mathbf{c_t}^{(\delta_C,\delta_S)} + \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{\delta_C}^{-\delta_S} \delta_C \mathbf{S}\bar{\mathbf{b}} \\ \mathbf{y_t} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{(\delta_C)}^{-\delta_S} \mathbf{S} \\ \mathbf{S} \end{bmatrix} \mathbf{b_t}$$

(2) We therefore have

$$\mathbf{z}_t^* = \begin{bmatrix} \mathbf{v}_t^{(\delta_C,\delta_S)} \\ \mathbf{a}_t \\ \mathbf{b}_t \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{(\delta_C)}^{-\delta_S} \mathbf{S} \\ \mathbf{A} \\ \mathbf{I}_{n_b} \end{bmatrix} \mathbf{b}_t$$

where $\mathbf{S}_p^* = \begin{bmatrix} \boldsymbol{\Phi}_p^* \\ \mathbf{I}_{n_b} \end{bmatrix}$ and $\boldsymbol{\Phi}_p^* = \begin{bmatrix} \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{(\delta_C)}^{-\delta_S} \mathbf{S} \\ \mathbf{A} \end{bmatrix} \in \mathbb{R}^{(p+n_a)\times n_b}$

(3) Hence, we use

$$\mathbf{G}_{h,p}^* = \left[(\mathbf{S}_p^*)'(\mathbf{W}_h^*)^{-1}\mathbf{S}_p^*\right]^{-1}(\mathbf{S}_p^*)'(\mathbf{W}_h^*)^{-1}$$

to obtain

$$\tilde{\mathbf{y}}_{t+h|t}^{\text{FLAP}} = \mathbf{S}\tilde{\mathbf{b}}_{t+h|t}^{\text{FLAP}} = \mathbf{S}\mathbf{G}_{h,p}^* \hat{\mathbf{z}}_{t+h|t}^*$$

2.4.8 Robustness: Using scaled errors

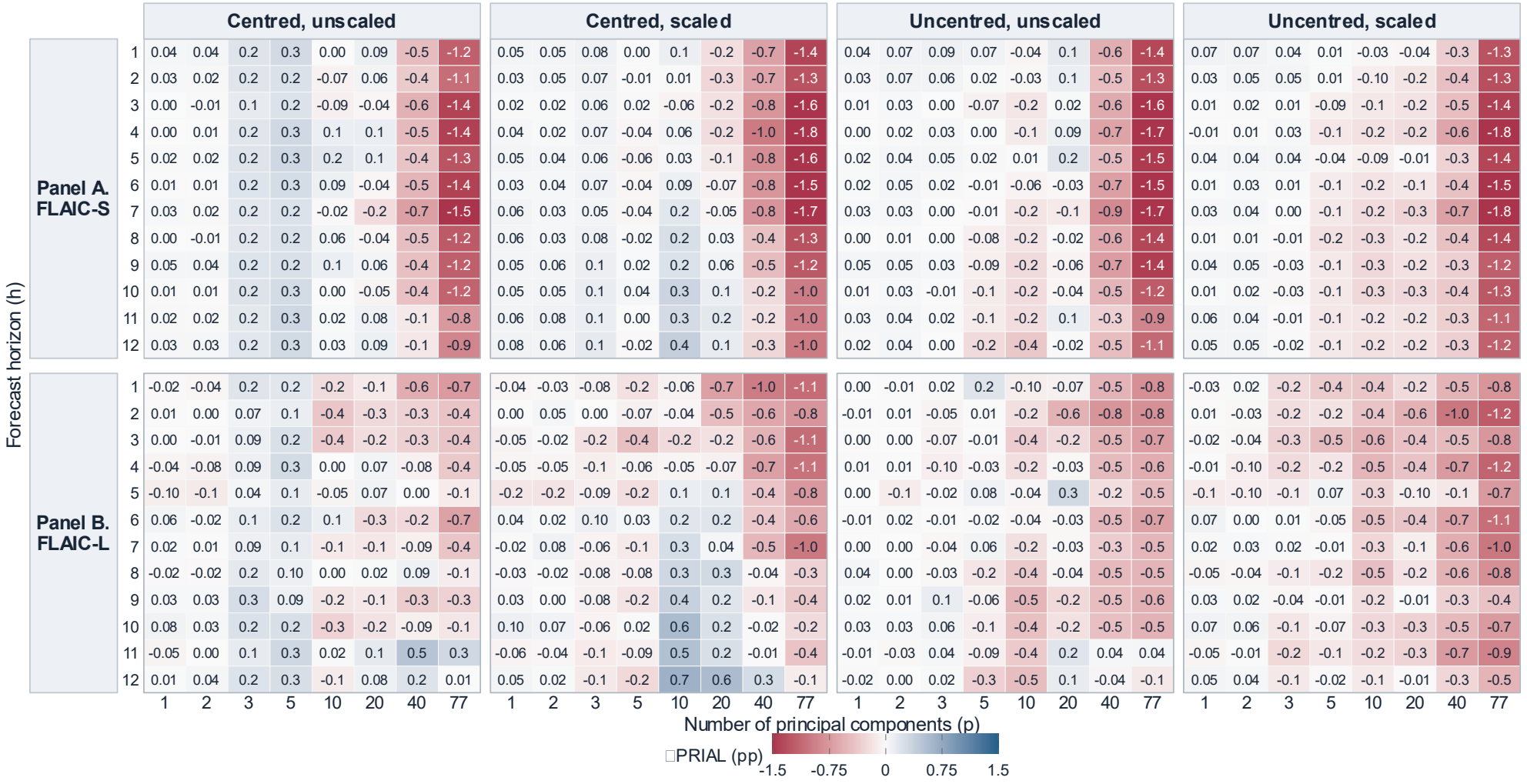


Figure: PRIAL of FLAIC relative to IComb: Average across 112 hierarchy series. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L. Uses **seasonal-naïve scaled errors** to compute MSSFE, the PRIAL

2.4.8 Robustness: Using scaled errors

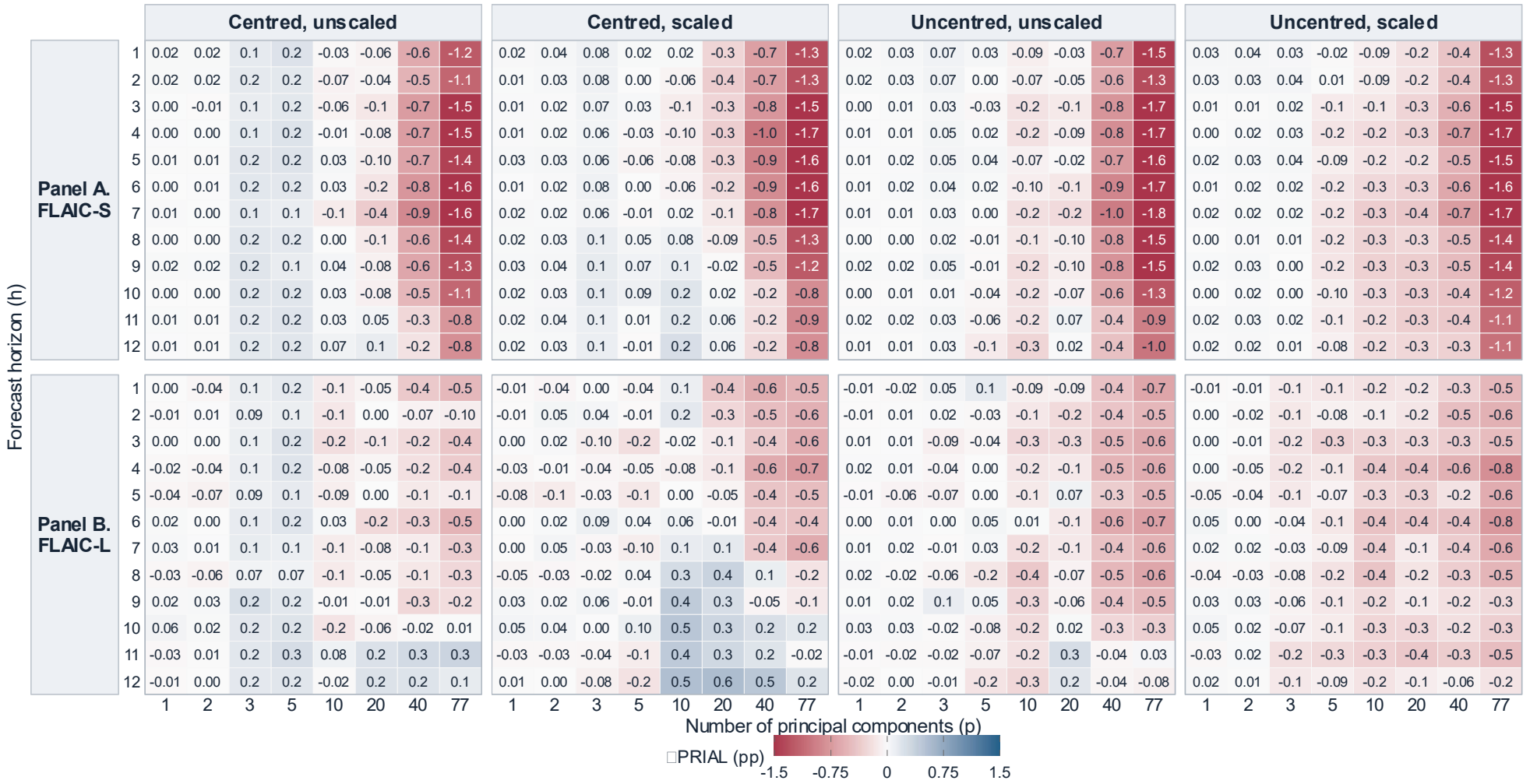


Figure: PRIAL of FLAIC relative to IComb: Average across 112 hierarchy series. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L. Uses **variance standardised scaled errors** to compute MSSFE, the PRIAL

MinT details

- Key benchmark: Minimum Trace (MinT) reconciliation (Wickramasuriya et al., 2019).

  Under unbiasedness of base forecasts:

$$\mathbf{G}_h^{\text{MinT}} = \left(\mathbf{S}'\mathbf{W}_h^{-1}\mathbf{S}\right)^{-1}\mathbf{S}'\mathbf{W}_h^{-1}$$

  where $\mathbf{W}_h$ is base forecast error covariance matrix

  - Unbiasedness constraint (which is constraint needed to preserve unbiasedness of reconciled forecasts if base forecasts are unbiased): $\mathbf{G}_h\mathbf{S} = \mathbf{I}_{n_b}$
  - Minimise trace of $\text{var}\left[\tilde{\mathbf{e}}_{t+h|t}\right]$ $\quad (\tilde{\mathbf{e}}_{t+h|t} = \mathbf{y}_{t+h} - \tilde{\mathbf{y}}_{t+h|t})$
  - $\mathbf{G}_h^{\text{MinT}} = \left(\mathbf{S}'\mathbf{W}_h^{-1}\mathbf{S}\right)^{-1}\mathbf{S}'\mathbf{W}_h^{-1}$, $\quad \mathbf{W}_h = \text{var}\left[\hat{\mathbf{e}}_{t+h|t}\right]$ $\quad (\hat{\mathbf{e}}_{t+h|t} = \mathbf{y}_{t+h} - \hat{\mathbf{y}}_{t+h|t})$
  - $\mathbf{W}_h$ estimated from in-sample forecast errors with $\mathbf{W}_h = \text{k}_\text{h}\widehat{\mathbf{W}}_1$

MinT details (cont.)

- MinT has OLS and WLS as special cases:
  - $\mathbf{W}_h$ is base forecast error covariance matrix
  - If we assume $\mathbf{W}_h = \mathrm{k_h}\mathbf{I}_n$, then we have OLS case
  - If we assume $\mathbf{W}_h = \mathrm{k_h}\,\mathrm{diag}(\widehat{\mathbf{W}}_1)$, then we have WLS (variance scaling) case (Hyndman et al., 2016)
  - If we assume $\mathbf{W}_h = \mathrm{k_h}\,\mathrm{diag}(\mathbf{S1})$, then we have WLS (structural scaling) case (Athanasopoulos et al., 2017)
  - MinT gives same answer as GLS even though different loss function being minimised (Wickramasuriya, 2021)
  - MinT and its special cases are considered projections as $\mathbf{SG}_h$ is a *projection* onto $\mathrm{span}(\mathbf{S})$. In these cases, $\mathbf{SG}_h$ is a projection matrix as it meets idempotence property: $(\mathbf{SG}_h)^2 = \mathbf{SG}_h$